

The Myth of the Monolithic Stone: An AI Multimodal Method for Modern Nephrolithiasis Management

Abstract

Kidney stones are routinely treated as monolithic objects summarized by a single CT attenuation value, yet most are **multicomponent** — a nidus laminated by shells of differing composition. We asked what a **standard acute emergency-department (ED) evaluation** — a single-energy non-contrast CT (NCCT) and the routine laboratory panel — can actually resolve about composition, reframing the problem around the **management decisions that diverge** rather than a composition taxonomy, since dual-energy CT is not available in routine ED care. In 319 laboratory-analyzed stones from 223 patients (**68% multicomponent**), under repeated patient-grouped cross-validation with patient-clustered bootstrap: **(Node A) the dissolvable decision — uric acid vs. non-uric-acid — was reliably resolved** (AUROC 0.76; high-sensitivity NPV 0.97; positive decision-curve net benefit across the plausible threshold range), the one acute decision that changes management. **(Node C1) single-energy CT could not detect multicomponent structure:** HU architecture (variance, entropy, modality, core-to-rim gradient) classified single- vs. multi-component stones at chance (AUROC **0.54, 95% CI 0.46–0.61**), and worse as the mixedness threshold tightened — the signature of partial-volume averaging erasing the laminated footprint. **(Node C2) a predominant-calcium-phosphate rule-in** — a metabolic flag for the alkaline-urine / distal-renal-tubular-acidosis / hyperparathyroid phenotype — reached AUROC **0.71 (0.64–0.78)** from CT plus labs. A systematic, pre-registered exclusion of **fourteen** feature families (HU calibration, deep learning, radiomic shape, density gradient, derived acid–base labs, opportunistic bone density, hepatic steatosis, sarcopenia, aortic calcification, and multicomponent architecture, among others) found **none improved** on this routine-data baseline. **The monolithic stone is a myth twice over:** stones are multicomponent, and single-energy CT cannot see it — so a single HU value is doubly misleading. The deployable contribution is a multimodal (imaging + laboratory) method targeting the decisions that actually diverge, and a negative-findings ledger that maps the boundary of routine data so others need not re-explore it. The residual calcium-oxalate-vs-phosphate gap — where it matters at all — is localized to stone-specific urinary analytes (spot oxalate) absent from routine panels.

1. Background and aim

Stone composition guides management. **Uric-acid stones dissolve** with urinary alkalinization and are often managed medically, whereas calcium and struvite stones are not. Identifying UA non-invasively, at presentation, could spare intervention and direct metabolic therapy. DECT addresses this but is concentrated in tertiary centers and is essentially **unavailable for routine ED use**, where the imaging is single-energy NCCT. Our explicit aim is a **real-world decision-support method using only routinely available data** — NCCT stone attenuation and routine ED labs — not a method that presumes specialized acquisition.

Scope and design principle. We deliberately restrict the predictor's inputs to what a **standard acute ED stone evaluation already generates**: a single-energy NCCT (stone attenuation, size, location; and any feature derivable from that volume) and the routine laboratory panel drawn at that encounter — complete blood count, basic metabolic panel, and urinalysis (from which pH is used) — with age and sex. No specialized acquisition

(DECT), no send-out or 24-hour metabolic panel, and no data unavailable at the index visit are used. This boundary is intentional and is central to the contribution: when an added feature fails to improve prediction within it, the result defines the **ceiling of routinely available data**, not an artifact of an incomplete work-up. The systematic feature search of Appendix A — and its uniformly negative result across fourteen families — should be read in that light: a map of what single-energy CT and standard ED chemistry can and cannot resolve, published so that others need not re-discover the same boundaries.

2. Methods

2.1 Cohort and data

Patients with surgically retrieved, laboratory-analyzed stones (FTIR/X-ray diffraction component percentages) and a pre-operative NCCT within 120 days were assembled from two sites, consolidated on a single medical-record identifier. Composition was mapped to parent classes (calcium oxalate, calcium phosphate, uric acid, struvite, cystine), and the binary endpoint was **UA-dominant vs. non-UA**. Routine labs at the time of CT were extracted: basic metabolic panel, urine pH, and demographics.

2.2 Imaging curation (non-contrast only)

From each study we selected a **non-contrast** CT series (contrast-enhanced and rendered/reformatted-only series excluded by DICOM contrast tags and acquisition type), preferring native axial reconstructions and falling back to non-contrast reformats where that was the only volume archived. HU is orientation-independent, so reformats are valid for attenuation measurement.

2.3 Stone segmentation and features

Stones were auto-segmented as high-attenuation components within the kidney/bladder region (TotalSegmentator organ masks), excluding skeleton and great-vessel calcification, with a size filter. Per stone we measured native-voxel HU statistics (**peak, mean, percentiles**), volume, and — in extended analyses — HU-distribution shape (skewness, kurtosis, number of histogram modes, normality) and morphology. The **locked model uses a deliberately lean feature set**, all routinely available in the ED:

stone peak and mean HU; urine pH; serum CO₂ (bicarbonate), chloride, anion gap (acid–base axis); BUN, creatinine, calcium, glucose; age, sex.

Monotonic constraints encode known chemistry (higher HU → less likely UA; higher urine pH → less likely UA).

2.4 Locked model and validation

The classifier is a regularized gradient-boosted tree (handles missing labs natively — important for real-world data). Validation is **repeated (25×) stratified, patient-level grouped 5-fold cross-validation** with averaged out-of-fold probabilities; **uncertainty is a patient-clustered bootstrap (1000×) 95% CI**. We report discrimination (AUROC), a high-sensitivity operating point (sensitivity/specificity/PPV/NPV), **calibration (Brier score)**, and **decision-curve analysis** (net benefit vs. treat-all/treat-none) as the clinical-utility readout.

2.5 Pre-specified alternatives (tested, not assumed)

To avoid over-engineering, four elaborations were pre-specified and tested against the lean model: (i) an **Hb-anchored internal HU calibration** (air + aortic-blood anchors); (ii) a **self-supervised 3D CNN** (frozen-encoder deep features); (iii) **HU-distribution-shape radiomics**; and (iv) a **larger multi-stone cohort** via volume $\leftrightarrow$ mass matching. A fifth elaboration — a **stone density gradient** (core-to-periphery HU ratio) — was tested in the composition-distribution extension (§2.6).

2.6 Extension: full composition distribution and calcium-subtype separation

Beyond the binary UA endpoint we built a **five-class composition model** (calcium oxalate, calcium phosphate, uric acid, struvite, other) that returns a calibrated probability distribution for point-of-care use, trained on the same routine inputs under the same patient-grouped protocol (20× repeated stratified grouped 5-fold). Two further questions were addressed.

Density gradient. Motivated by the observation that calcium-oxalate stones often retain a dense core whereas calcium-phosphate may taper, we computed the **core-to-periphery HU ratio** with shells defined by **physical depth** from the stone surface — a spacing-aware Euclidean distance transform, split at the 33rd/67th depth percentiles, summarized by **median** HU (robust to the partial-volume rim). The feature is **gated** to stones large enough to define concentric shells (≥ 150 voxels total, ≥ 15 per shell); smaller/thinner-slice stones return missing rather than a slice-thickness artifact. The ratio was re-extracted for the labeled cohort by re-segmenting each study, and its stability was audited **by reconstruction kernel** (soft vs. sharp) and by composition.

Calcium-oxalate-vs-phosphate head. A **dedicated binary head** (positive class = calcium phosphate) was compared across nested feature sets — HU only; HU + gradient; HU + routine labs; HU + labs + gradient — under **40× repeated patient-grouped 5-fold CV**, with patient-clustered bootstrap CIs on AUC and on the **paired** feature-set deltas. The operating threshold was set from cross-validated out-of-fold P(CaP) to a **high-sensitivity** point for use as a calcium-phosphate work-up screen. The head is wired as a **cascade**: it re-partitions the combined calcium-oxalate + calcium-phosphate probability mass in the five-class distribution by its P(CaP|calcium), leaving the non-calcium classes and the total unchanged.

3. Results

3.1 Cohort

319 clean-label stones from 223 patients; **UA prevalence 11%**. Composition was dominated by calcium phosphate and calcium oxalate, with uric acid and struvite minorities and cystine rare.

3.2 Discrimination and operating characteristics

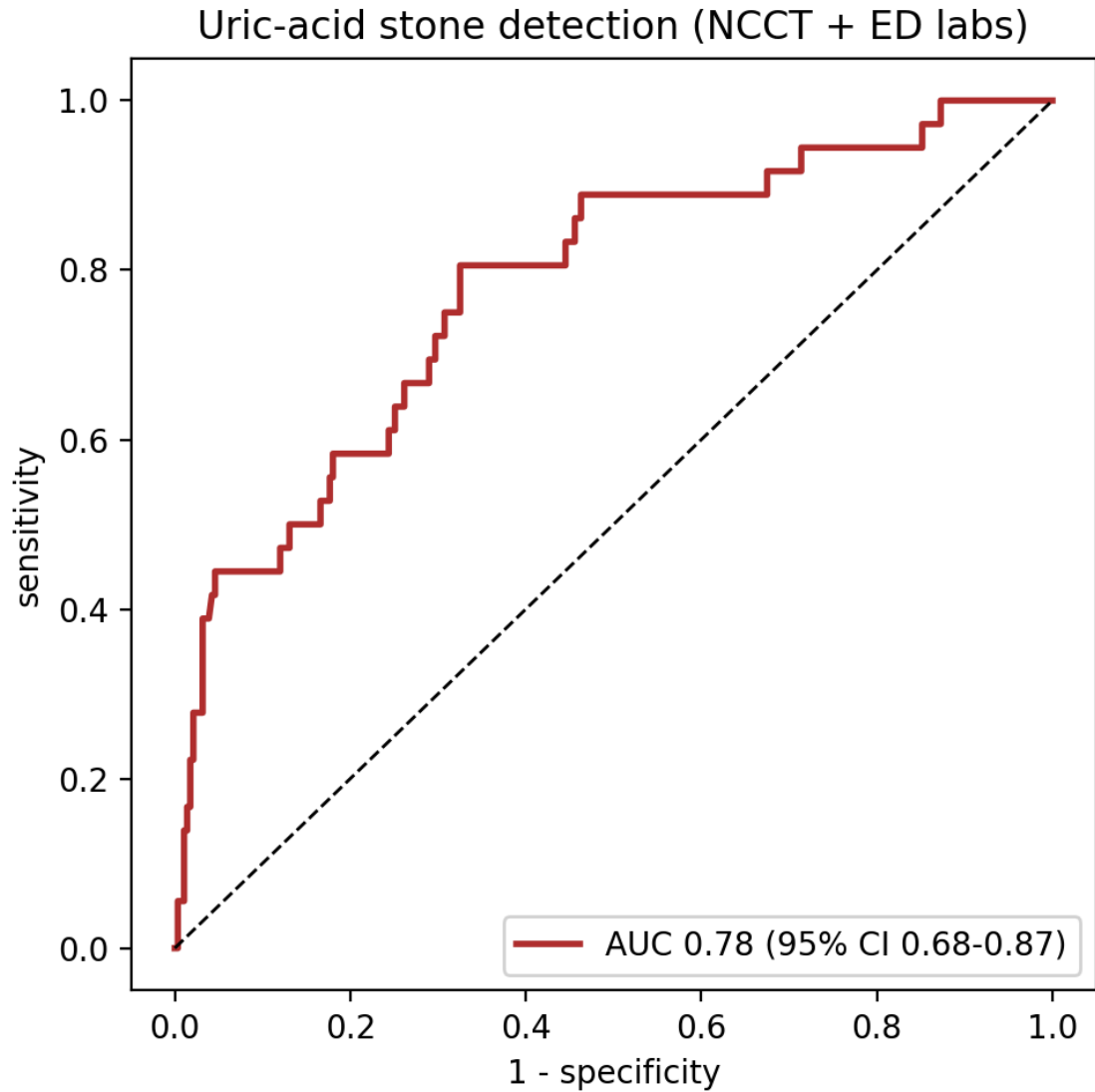


Figure 1. UA-stone detection (NCCT + ED labs). AUROC 0.781 (95% CI 0.677–0.867). At a high-sensitivity threshold: **sensitivity 0.92 (0.81–1.00)**, specificity 0.33 (0.25–0.39), PPV 0.15, **NPV 0.97** — a low predicted probability reliably excludes UA. Because UA prevalence is low, the model is best used as a **calibrated probability that updates clinical suspicion**, not a hard classifier.

3.3 Clinical utility

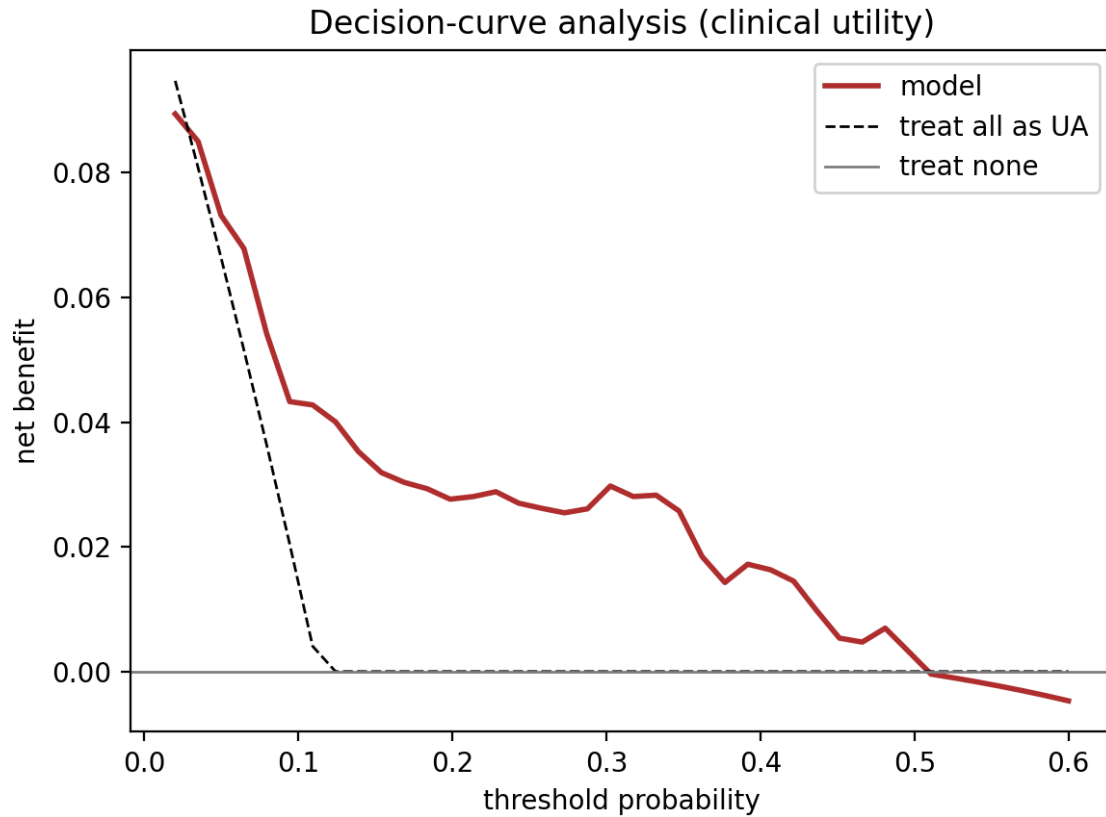


Figure 2. Decision-curve analysis. The model yields **positive net benefit across the entire clinically plausible threshold range (~10–50%)**, exceeding both "treat all as UA" (which loses benefit beyond ~12%) and "treat none." This is the key real-world result: for any clinician whose action threshold for "consider UA" falls in this range, the model adds value over default strategies.

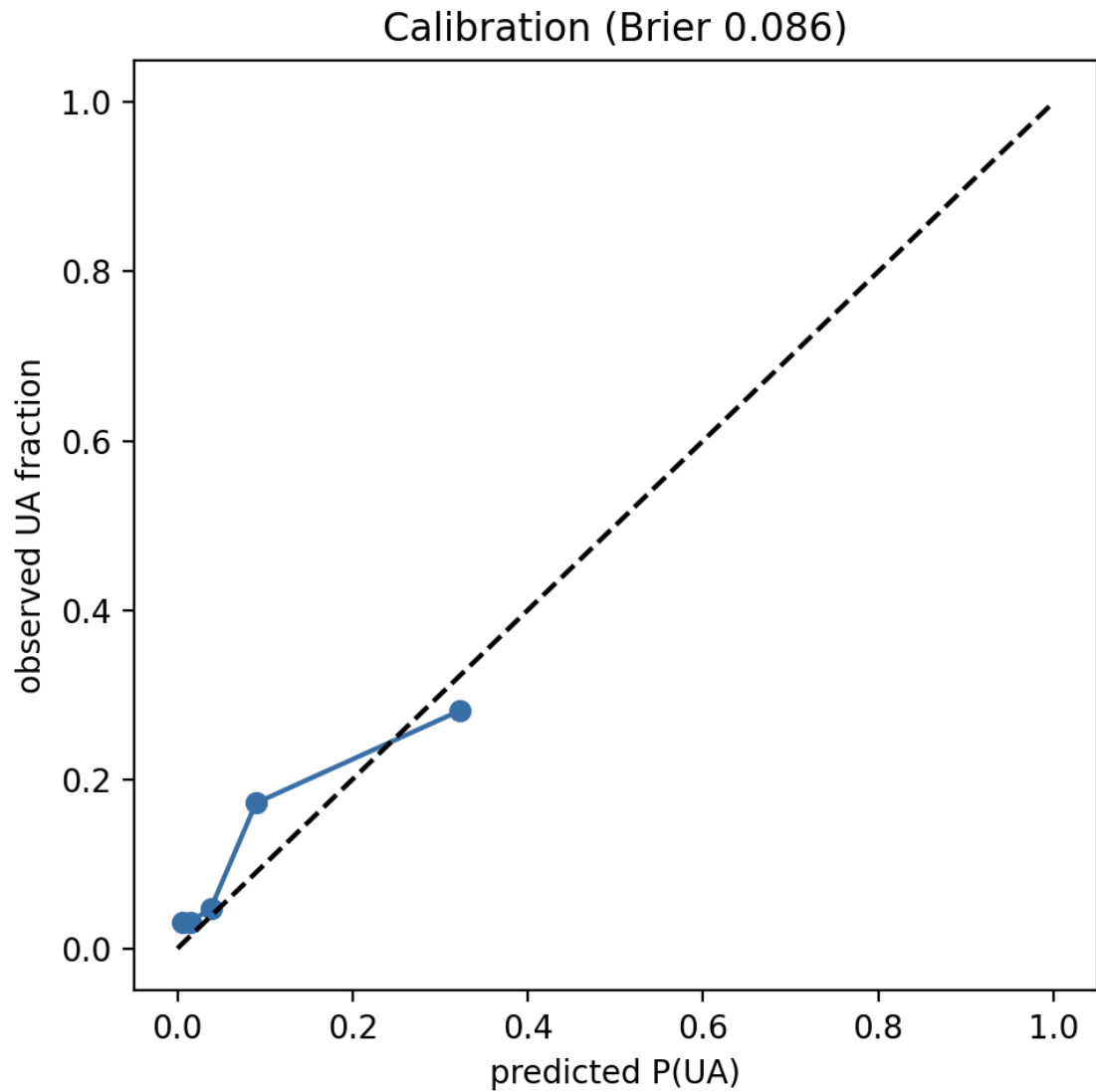


Figure 3. Calibration. Predicted UA probabilities track observed frequencies (**Brier 0.086**), supporting use of the continuous score for shared decision-making.

3.4 What drives the prediction

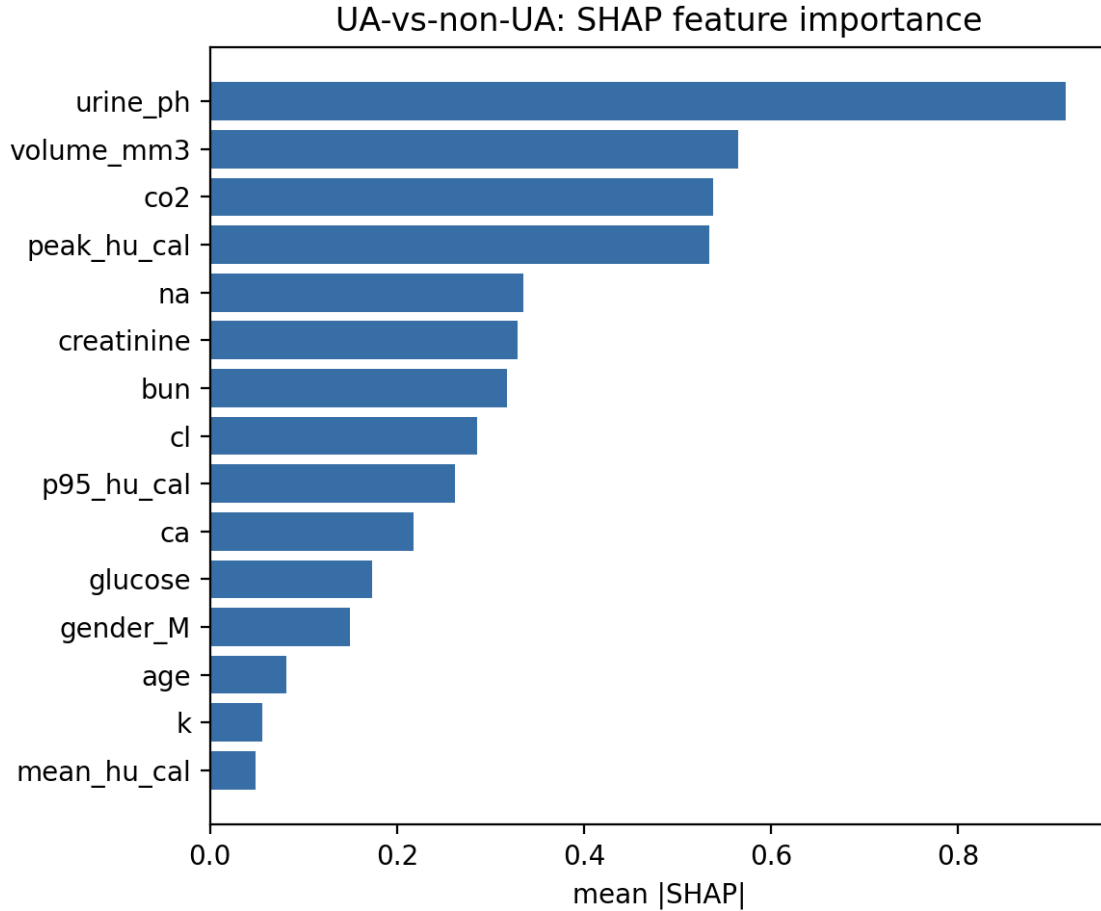


Figure 4. Feature importance (SHAP). The model leans on **urine pH** and **stone HU**, with the **acid–base axis (bicarbonate, chloride, anion gap)** prominent — consistent with stone chemistry (UA forms in persistently acidic urine; the normal-anion-gap/alkaline-urine pattern of distal renal tubular acidosis favors calcium-phosphate stones). Directions are chemically correct (low pH and low HU push toward UA).

3.5 Elaborations that did not help

None of the four pre-specified alternatives improved on the lean model: Hb-anchored calibration gave no reduction in same-composition HU variance (the scanner offset, ~10 HU, is dwarfed by ~220 HU of biological spread); the 3D CNN underperformed and degraded fusion (deep features did not cluster by composition); HU-distribution-shape features **reduced** cross-validated AUC; and the multi-stone N-lift (131 → 319) was **neutral**. The density-gradient extension (§3.6) was likewise rejected. The single-energy composition signal is captured by attenuation and chemistry, and added complexity overfit at this scale.

3.6 Extension results: a biological null and a modest labs-driven head

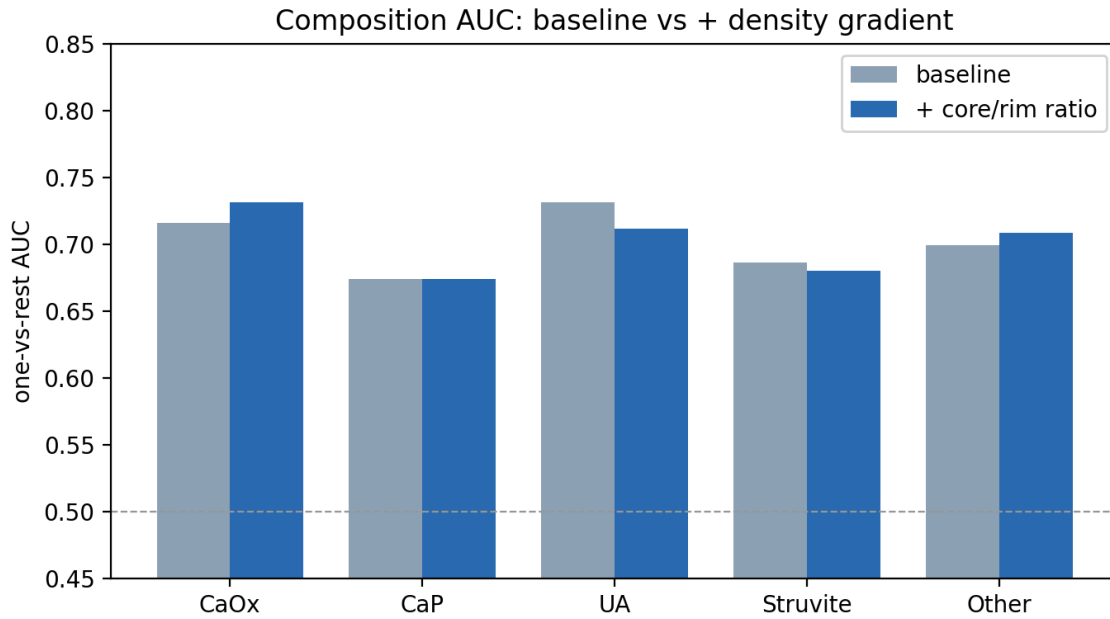


Figure 5. Density gradient — per-class AUC, baseline vs. + core/periphery ratio. The **density gradient was measurable and stable but non-discriminating for the calcium subtypes**. It was computable on **90% of cohort stones** and stable across reconstruction kernels (coefficient of variation ≈ 0.4), yet the mean core-to-periphery ratio was **2.86 for calcium oxalate vs. 2.80 for calcium phosphate** — a 0.06 separation against a within-group SD of ≈ 1.2 . Added to the five-class model it slightly **lowered** macro-F1 ($0.317 \rightarrow 0.296$) and ranked **13th of 17** features by SHAP; the only real signal it carried was the greater homogeneity of **uric-acid** stones (ratio 2.12), already captured by HU and urine pH. The hypothesized oxalate-dense-core / phosphate-taper contrast is therefore **not resolved by single-energy CT** — a clean negative, because the feature was well-measured rather than noisy.

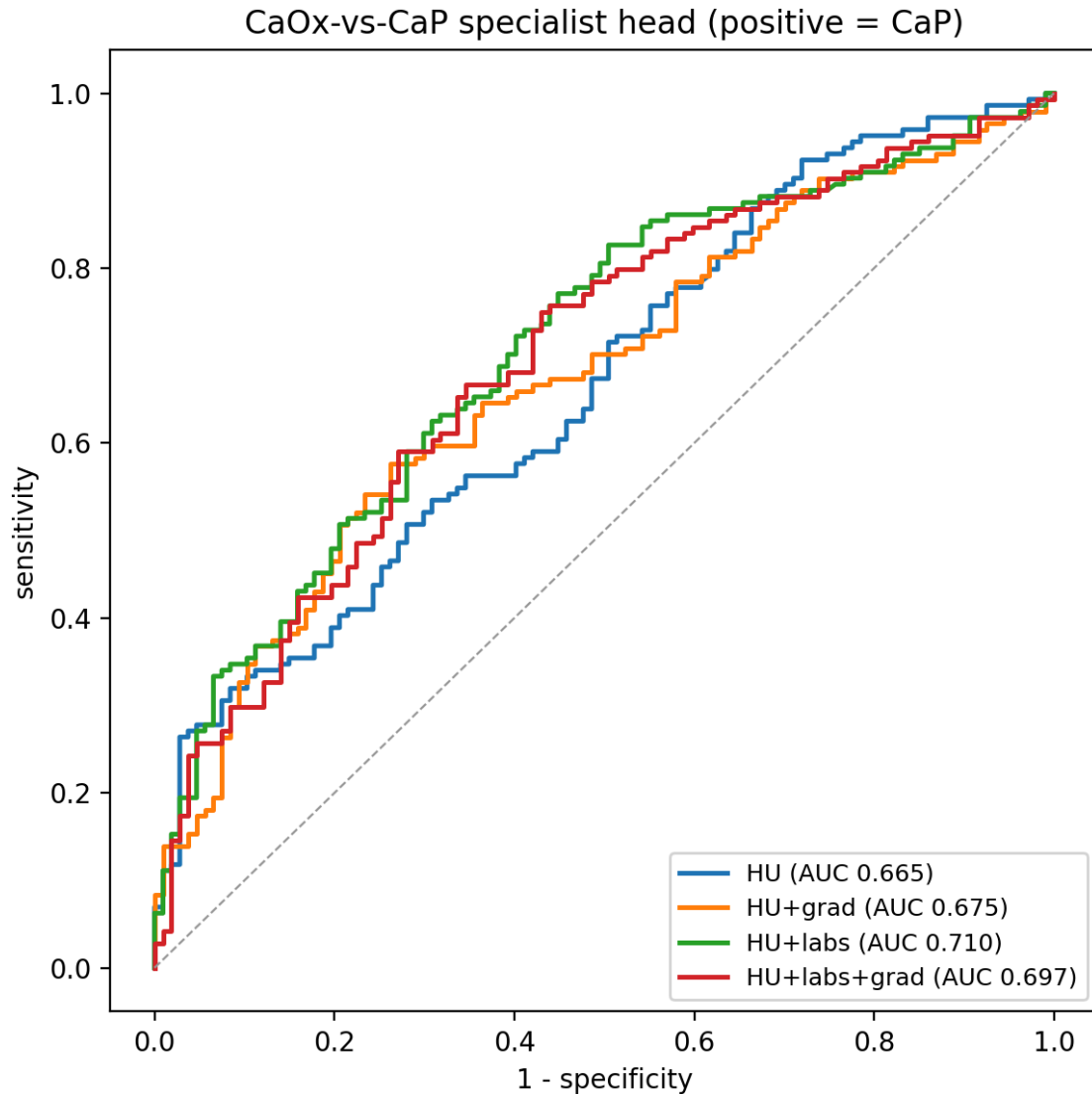


Figure 6. Calcium-oxalate-vs-phosphate head — ROC by feature set (positive = CaP). A targeted head added modest value, but from labs, not imaging. Cross-validated AUC was 0.665 (HU), 0.675 (HU + gradient), **0.710 (HU + labs; best)**, and 0.697 (HU + labs + gradient); the gradient's apparent benefit over HU alone **vanished against the stronger baseline** (paired $\Delta -0.013$, 95% CI -0.034 to $+0.007$, $P[\Delta > 0] = 0.11$). The deployed head therefore uses **HU + routine labs without the gradient**, reaching **AUROC 0.710 (95% CI 0.637–0.778)** and exceeding the five-class model's one-vs-rest calcium-phosphate AUC (0.674). Its discrimination is driven by **urine pH and the acid-base panel**, consistent with calcium-phosphate biology (alkaline urine, distal renal tubular acidosis, hypercalcemia).

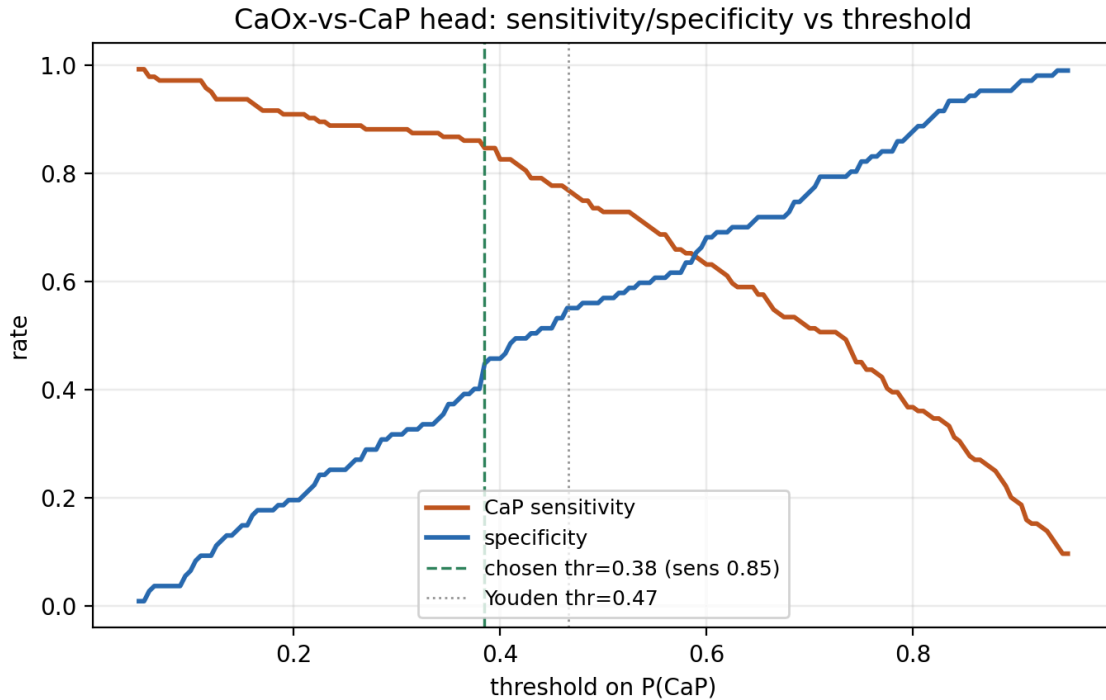


Figure 7. CaP screen operating point. Because the AUROC ceiling is ~ 0.71 , sensitivity trades steeply against specificity. The screen is set to a **high-sensitivity** point chosen from cross-validated out-of-fold predictions — **target sensitivity 0.85: threshold 0.385, specificity 0.45, NPV 0.70** (vs. 0.90 sensitivity $\rightarrow$ specificity 0.23). It functions as a calcium-phosphate **work-up prompt** (urine pH, serum calcium/PTH, distal-RTA evaluation; caution with potassium citrate, which raises urine pH and can worsen calcium-phosphate stones), surfaced only when calcium is a genuine contender — explicitly a rule-in screen (PPV 0.68), not a confirmation.

3.7 Decision-node performance

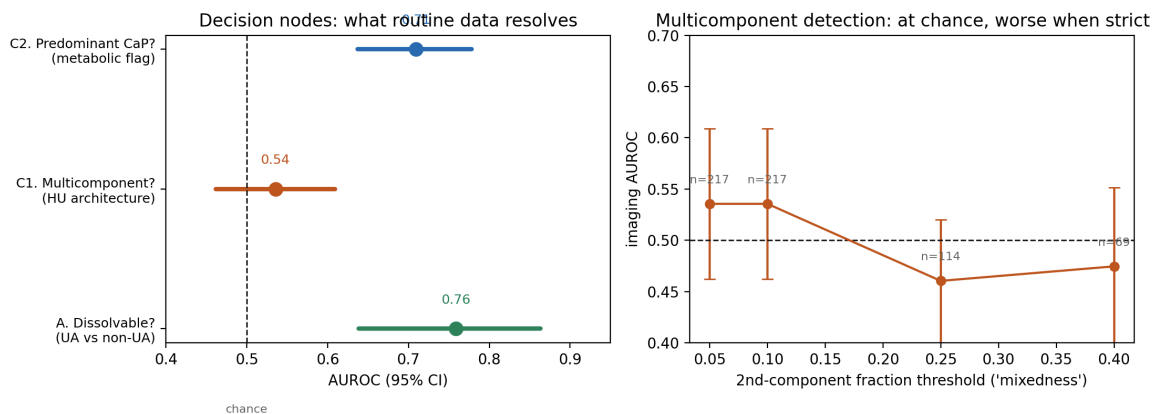


Figure 8. Reframing around management decisions. Recasting the problem as the decisions that diverge — rather than a composition taxonomy — clarifies both what routine data delivers and what it cannot. **Node A (dissolvable?)** — uric acid vs. non-UA, the one acute decision that changes management (medical dissolution vs. intervention) — was resolved at **AUROC 0.76 (95% CI 0.64–0.86)** from CT + labs. (Node B,

infection/struvite, is a clinical diagnosis — alkaline turbid malodorous urine, recurrent UTI, sepsis — and is not a modeling target.) **Node C1 (multicomponent?)** — whether a stone is single- vs. multi-component, a coarser and better-balanced task (68% multicomponent) that, per the lamination mechanism, should leave an HU-architecture footprint even without resolving composition — was **not detectable**: imaging AUROC **0.54 (0.46–0.61)** at the loosest definition, falling *below* chance as the second-component threshold tightened (0.46–0.48 at ≥ 25 –40%). That is the opposite of a resolvable signal (a cleaner two-phase stone should be *easier*), and the expected consequence of partial-volume averaging at clinical resolution: the core-to-rim and HU-shape features ranked on top by SHAP yet carried no discrimination. **Node C2 (predominant CaP?)** — a calcium-phosphate rule-in flagging the metabolic phenotype — reached **AUROC 0.71 (0.64–0.78)** (Youden: sensitivity 0.77, specificity 0.55), operationalized as the high-sensitivity work-up screen and the nephrocalcinosis flag (§3.6, R3). The through-line: routine single-energy data resolves the dissolvable decision that matters and supports a modest metabolic flag, but is blind to the multicomponent architecture that in fact characterizes most stones — the monolithic-stone assumption fails both because stones are mixed and because the imaging cannot see that they are.

4. Discussion

The clinically actionable target in single-energy CT is **UA vs. non-UA**, and a model using only **routine ED data** discriminates it with an AUROC of ~ 0.78 and, importantly, **positive net clinical benefit** across realistic decision thresholds. The high NPV makes it well suited to **ruling out** the dissolvable stone and to **flagging** candidates for alkalinization or metabolic work-up. Calcium-oxalate vs. calcium-phosphate separation is poor from **imaging** — the density-gradient extension confirms that single-energy attenuation does not resolve the two subtypes structurally, even with a well-measured, kernel-stable feature — yet routine **acid–base labs** recover a modest, biologically-grounded separation (a dedicated head at AUROC 0.71), deployed as a calibrated refinement of the calcium probabilities and a high-sensitivity work-up screen rather than a replacement for stone analysis. This is exactly where DECT, when available, retains an advantage; our method is complementary, targeting the decisions single-energy data **and routine labs** can support.

The chief value is **deployability and honesty**: every input is already collected on ED stone patients, and four tempting elaborations were tested and rejected, guarding against the over-fitting that inflates many radiomics reports.

Limitations. Retrospective, two-site, modest UA count ($n=36$) → wide CI; spot (not 24-hour) urine chemistry; no explicit diabetes/HbA1c field (entered via glucose); rank-based multi-stone matching; and the need for **prospective, external validation**. The calcium-subtype extension is **modest and preliminary**: the head is trained on a 251-stone calcium subset, reaches only AUROC 0.71, and is deployed as a **rule-in screen** (PPV 0.68), not a confirmation; the density-gradient negative is established in this single-energy cohort and does not speak to dual-energy or micro-CT. The model is decision-support, not a replacement for stone analysis.

A note on the urinalysis, and the bridge to prospective work. Within our routine-data scope, one input was captured only in part: from the urinalysis we used **urine pH alone**. A standard ED urinalysis additionally reports specific gravity (a concentration reference for spot analytes) and, on microscopy, **crystalluria** — calcium-oxalate (envelope/dumbbell), uric-acid, and struvite ("coffin-lid") morphologies — which carry composition-relevant information we did not have. Crystalluria is insensitive and operator-dependent, so it is

unlikely to be the discriminator on its own, but it is genuinely part of a standard visit and a prospective design should record it. More consequentially, the uniformly negative feature search (Appendix A) indicates that the residual **calcium-oxalate-vs-phosphate** limit will not yield to further imaging or routine chemistry — both are saturated — and instead points to a single **stone-specific urinary analyte** absent from routine panels: **spot urinary oxalate referenced to creatinine**. Oxalate is the swing driver of calcium-oxalate supersaturation (urinary calcium far exceeds oxalate, so supersaturation tracks oxalate), it is inexpensively measured enzymatically (oxalate oxidase), and an elevated ratio would specifically incriminate calcium oxalate — an *asymmetric, rule-in* marker (a normal value would not exclude calcium oxalate, since many such formers are normo-oxaluric). This defines the shortest prospective extension of the present work: adding reflex spot oxalate/creatinine to the routine-data model, evaluated on the calcium stones this model leaves indeterminate.

5. Conclusion

Attenuation on routine non-contrast CT, combined with routine ED labs centered on urine pH and the acid–base axis, identifies uric-acid stones with clinically useful, well-calibrated performance and net benefit across realistic thresholds — using data the emergency department already has, without dual-energy CT. The deliberately lean, interpretable model outperformed every elaboration tested (Appendix A): calibration, deep learning, radiomic shape, larger cohort, stone density gradient, derived acid–base labs, eGFR, opportunistic bone-density, body composition, vascular calcification, and multicomponent architecture. Where a finer distinction is possible at all — calcium oxalate vs. phosphate — it comes from **routine acid–base chemistry rather than from imaging**, and is delivered as a calibrated refinement and high-sensitivity work-up screen; the one genuinely orthogonal imaging finding (nephrocalcinosis → calcium phosphate) is mechanistically real but, at this sample size, supportable only as a radiologic flag, not a learned feature. The result is a credible, honest basis for prospective evaluation as real-world decision support in stone disease.

Appendix A. Feature-search ledger (what else was tested, and rejected)

Beyond the four pre-specified elaborations (§2.5) and the density gradient (§2.6/§3.6), we continued searching for any additional signal extractable from the **routine labs** or from the **non-contrast CT beyond the stone itself**. Everything below was evaluated under the same patient-grouped repeated cross-validation with patient-clustered bootstrap CIs, on the CaOx-vs-CaP head (the weak spot) and the five-class model. All were rejected; the pattern is informative.

A.1 Derived acid–base / renal labs (no new measurements). An explicit normal-anion-gap (hyperchloremic) metabolic-acidosis composite — the distal-RTA / GI-bicarbonate-loss signature — plus a hyperchloremia index and CKD-EPI eGFR. None improved discrimination: on the CaOx-vs-CaP head, HU+labs 0.710 → +acid-base 0.709 ($\Delta -0.000$, 95% CI -0.009 to $+0.008$) and +eGFR 0.700 ($\Delta -0.009$). SHAP attended to eGFR (rank 7/19) yet out-of-fold AUC was flat — a clean demonstration that **the acid–base axis is already saturated by the raw bicarbonate/chloride/anion-gap values** and eGFR is redundant with creatinine/age/sex. SHAP attention is not predictive value, which is why adjudication is on grouped-CV AUC, not feature importance.

A.2 Imaging beyond the stone. From the same TotalSegmentator pass we extracted two patient-level features: **nephrocalcinosis** (parenchymal calcific foci within the eroded renal cortex — collecting-system stone and

bone/vessel calcification removed, with a per-focus size cap to exclude staghorn bleed-through) and **opportunistic vertebral BMD** (trabecular-core L1–L3 median HU). - The **biology was the cleanest of the entire search**: mean nephrocalcinosis fraction 0.0074 in calcium-phosphate vs 0.0000 in calcium-oxalate stones ($\approx 10\times$), with prevalence rising monotonically CaOx 21% → CaP 33% → struvite 48% — exactly the distal-RTA / hyperparathyroidism / infection pathways. - On the full cohort, +nephrocalcinosis+BMD produced the only nominally significant gain of the whole search (CaOx-vs-CaP +0.038, 95% CI +0.006 to +0.074). **It did not survive verification**. Complete-case analysis (restricting to patients with BMD actually measured, removing the 35% missing) collapsed the interval across zero (+0.024, 95% CI −0.021 to +0.069); dropping the 15 heavy-nephrocalcinosis outliers reduced the nephrocalcinosis effect from +0.013 to +0.004. The apparent gain was carried by a handful of high-burden cases plus a BMD contribution that vanished complete-case (BMD's by-composition means were essentially flat — 187 vs 194 HU — i.e. it re-encodes age). - **BMD was missing for 35%** of patients: stone-protocol CT is centered on the urinary tract, so the lumbar vertebral bodies are frequently clipped (total z-coverage was similar for BMD-present and BMD-missing scans — the difference is FOV centering and series type, not scan length). Importantly, missingness was **not** associated with composition (CaP rate 0.57 present vs 0.58 missing), so BMD neither leaked through its missingness nor helped.

A.3 Body composition (hepatic steatosis, sarcopenia). A microbiome-motivated hypothesis was tested: gut dysbiosis simultaneously raises enteric oxalate absorption (loss of *Oxalobacter*-mediated degradation → calcium-oxalate) and drives hepatic fat (NAFLD), so **fatty liver should be more prevalent in calcium-oxalate than calcium-phosphate** — a pathway orthogonal to urine pH. We measured hepatic steatosis (liver–spleen median HU) and muscle quality (paraspinal+psoas mean HU, a height-free myosteatosis marker), with **99% coverage** (liver/spleen and muscle are reliably in the urinary-tract FOV, unlike the lumbar spine). The hypothesis was **not supported, with full power**: calcium-oxalate vs calcium-phosphate steatosis rates were 13% vs 17% (liver–spleen +2.4 vs +1.4 HU — calcium-phosphate marginally *fattier*, opposite the prediction; Mann–Whitney $p=0.71$). The fatty-liver signal instead tracked **uric-acid** stones (steatosis 25%, liver–spleen −2.3) — the metabolic-syndrome/insulin-resistance phenotype already captured by urine pH and glucose. Added to the models the features **degraded** performance: the CaOx-vs-CaP head fell from 0.710 to 0.687 with muscle ($\Delta -0.023$, 95% CI −0.049 to +0.001) and the five-class macro-F1 from 0.317 to 0.294. Notably muscle *volume* drew high SHAP attention (rank 5/20) while actively *hurting* out-of-fold AUC — body-size confounding the model latches onto in-sample but that does not generalize, the clearest illustration in the ledger that **feature importance is not predictive value**. (A mechanistically sensible sidelight — struvite had the lowest muscle HU, the frail/recurrent-infection phenotype — was not actionable at $n=27$.)

A.4 Vascular and chronic-infection markers. Aortic wall calcification was tested as a readout of calcium-phosphate / mineral metabolism (CKD-MBD, high $\text{Ca}\times\text{PO}_4$) — a phosphate axis the labs do not measure (no serum phosphate/PTH), hypothesized higher in calcium-phosphate than calcium-oxalate. An initial 130-HU shell over-detected ($\approx 90\%$ prevalence in an age-47 cohort = partial volume), so the feature was re-extracted Agatston-style (focus peak ≥ 200 HU, 4–200 mm³ size window, density-weighted score, broadened bone exclusion), dropping prevalence to a plausible 78% and cleanly separating genuinely calcified patients from noise. The clean feature was **null and, if anything, reversed**: calcium-oxalate had marginally *more* aortic calcium than calcium-phosphate (median Agatston 330 vs 249, $p=0.31$), the **age-adjusted coefficient was −0.10 (95% CI −0.41 to +0.21)**, and the CaOx-vs-CaP head was unmoved ($\Delta +0.001$). Vascular calcification

indexes the hypercalciuria/calcium-overload phenotype (calcium-oxalate) as much as the phosphate axis, netting to nothing. Bladder wall fraction / volume and prostate calcification (exploratory struvite markers) produced an apparent but **spurious** gain: prostate calcification ranked #1 of 22 by SHAP yet is simply a **sex indicator** (present in 79% of males, 1% of females), drove the entire effect, was non-significant (95% CI crossing zero), redundant with the sex already modeled, and *lowered* five-class struvite AUC (0.686 $\rightarrow$ 0.626) — the project's sharpest example that feature importance is not predictive value.

A.5 Multicomponent detection — an exhaustive negative (the monolithic-stone test). Because dual stones laminate — a nidus shelled by a second phase of distinct composition and density — we asked whether *any* routinely available channel could flag single- vs. multi-component stones **without** resolving composition, a coarser and better-balanced task (68% multicomponent) than subtyping. The multicomponent label was swept over the second-component fraction ($\geq 5\%$ to $\geq 25\%$), and **every feature family of this study was tested against it** — stone HU, HU-shape radiomics, density gradient, Hb-calibrated HU, self-supervised **CNN deep features (64-dim)**, routine labs/urine, derived acid-base/eGFR, demographics, nephrocalcinosis, bone density, hepatic steatosis, sarcopenia, aortic calcification, and bladder/prostate markers — plus a dedicated **peak-deconvolution** family (per-stone 1- vs 2-Gaussian mixture fits yielding peak separation, Sarle's bimodality coefficient, skewness, and the 2-vs-1 BIC evidence), under patient-grouped CV with clustered bootstrap (Figure S4).

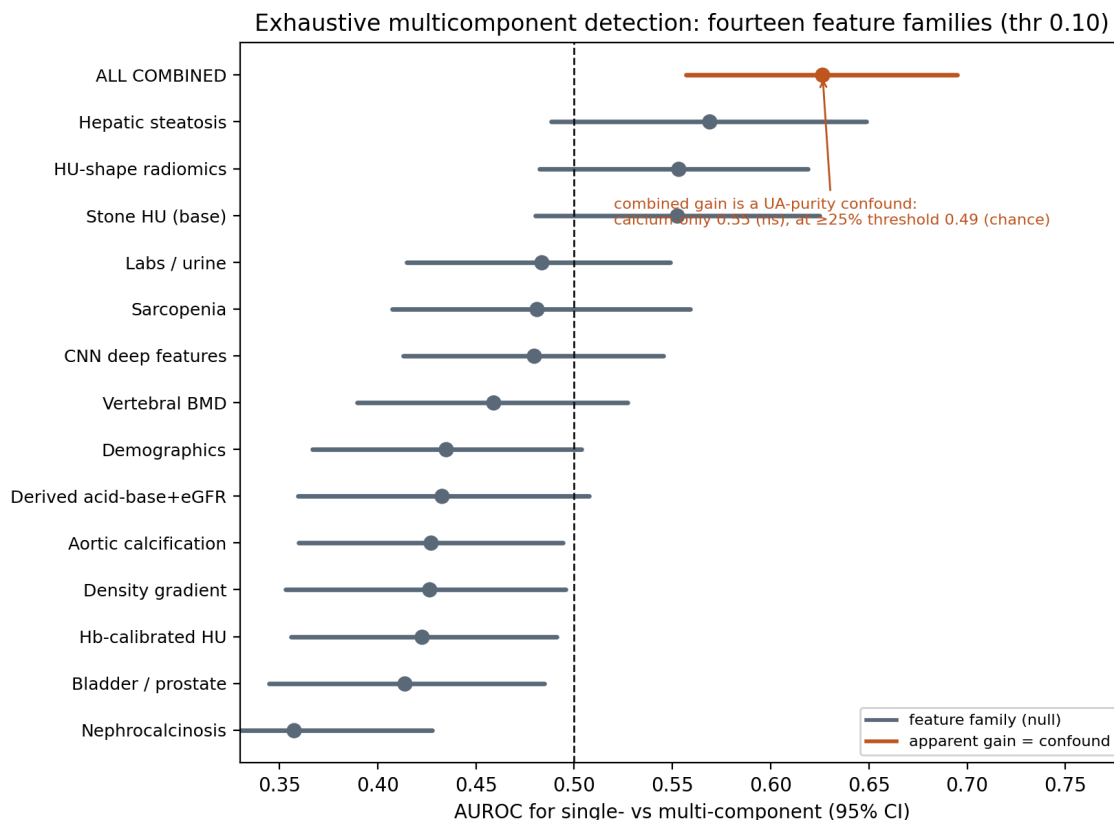


Figure S4. Exhaustive multicomponent detection. Every feature family's 95% CI crosses chance (several sit below it). **The one apparent gain — a 115-feature combined model, AUROC 0.626 (0.557–0.695) at the loosest threshold — is a uric-acid-purity confound, not multicomponent detection: UA stones are 39%**

multicomponent vs. 72% for non-UA, so "predict mixed" was largely "predict not-UA," which the model already does (Node A). Restricting to calcium stones (removing the UA axis) collapsed the combined model to 0.549 (non-significant); a clean $\geq 25\%$ mixedness threshold collapsed it to 0.491 (chance) — the two orthogonal controls that expose the confound. The **deconvolution** family, the most direct attempt (explicitly fitting two overlapping peaks and measuring separation and skewness), was itself at chance (0.53 size-robust), *below* chance at the clean threshold (0.375), added nothing to HU-shape (0.526 $\rightarrow$ 0.541), and was matched by a stone-volume-only baseline — what little varied was stone size, not bimodality. At clinical slice thickness (≈ 3 mm), partial-volume averaging erases the two-peak structure. **The multicomponent architecture that characterizes most stones is undetectable by any channel a standard ED visit provides** — the monolithic-stone assumption fails, and single-energy CT cannot correct it.

A.6 Disposition. Nephrocalcinosis is retained as a **deterministic radiologic flag** (substantial parenchymal calcification $\rightarrow$ favor calcium phosphate; evaluate for distal RTA / hyperparathyroidism / medullary sponge kidney), surfaced in the report when foci are auto-detected — not as a model input it is too sparse to support. BMD, body composition, vascular calcification, HU architecture, and the derived labs are not used. Across the full ledger — calibration, deep learning, HU-shape radiomics, larger cohort, density gradient, derived acid-base, eGFR, BMD, hepatic steatosis, sarcopenia, aortic calcification, bladder/prostate markers, and multicomponent architecture — single-energy prediction proved **saturated by stone attenuation and urine/acid-base chemistry**, and the one orthogonal imaging signal (nephrocalcinosis $\rightarrow$ calcium phosphate) is mechanistically real yet, at $n=144$ calcium-phosphate stones, confirmable only as a flag. The most likely path to the remaining calcium-subtype gap — where it matters — is a **stone-specific urinary analyte (spot oxalate/creatinine)** absent from routine panels, not further mining of the single-energy CT.